THE (ALMOST) IMPOSSIBLE PUZZLE

Abstract. Tom, Jerry and Spike play the following game. Tom generates a random binary code of length n, which he sends to Spike along with a *secret number* k between 0 and $n-1$. Spike flips exactly one bit in the code and then sends only the modified binary code to Jerry. Is there a pre-established strategy for Spike and Jerry such that Jerry can determine the *secret number* k solely by looking at the code sent by Spike?

1. Definitions

Let $\mathrm{Fin}\, n := \{0, \dots, n-1\}$, $n \in \mathbb{N}$ ($\mathrm{Fin}\, 0 := \emptyset$). $\mathbb{Z}_2 := \{0, 1\}$ is the commutative ring of integers modulo 2.

def:Z2^n **Definition 1.1.** $\mathbb{Z}_2^n := \mathbb{Z}_2 \times \dots \times \mathbb{Z}_2$. $\mathbb{Z}_2^n$ can be seen as the set of all functions from $\mathrm{Fin}\, n$ to $\mathbb{Z}_2$. In this file, we consider $\mathbb{Z}_2^n$ as a module over $\mathbb{Z}_2$.

def:e **Definition 1.2.** For $i \in \mathrm{Fin}\, n$, let $\mathrm{e}_i \in \mathbb{Z}_2^n$ be the canonical vector with 1 on the i-th position and zeros elsewhere

e (e) $$\mathrm{e}_i := (0, \dots, 0, 1, 0, \dots, 0),$$

e_i can be also seen as the function defined on $\mathrm{Fin}\, n$, supported at i, with value 1 there, and 0 elsewhere.

Note that $b + \mathrm{e}_i$ is the code b with the i-th bit flipped, for every $b \in \mathbb{Z}_2^n$ and $i \in \mathrm{Fin}\, n$.

def:SF **Definition 1.3.** We say that $f : \mathbb{Z}_2^n \to \mathrm{Fin}\, n$ is a **strategy function**, if

SF (SF) $$\forall\, b \in \mathbb{Z}_2^n, k \in \mathrm{Fin}\, n, \exists\, i \in \mathrm{Fin}\, n, f(b + \mathrm{e}_i) = k.$$

We say that the puzzle **has a strategy** for n if

$$\exists\, f : \mathbb{Z}_2^n \to \mathrm{Fin}\, n,\ (\mathrm{SF}).$$

def:d **Definition 1.4.** Let $\delta : \mathrm{Fin}\, n \times \mathrm{Fin}\, n \to \mathbb{N}$ be given by

d (d) $$\delta(i, j) \;=\; \text{if } i = j \text{ then } 1 \text{ else } 0.$$

See the Mathlib definition if-then-else.

2. Main result

In this section, we characterize the values of n for which the puzzle has a strategy and we present a sketch of the proof. The details of the proof are given in the next sections and are adapted for formalization in Lean.

Theorem *The puzzle has a strategy for* $n \in \mathbb{N}^*$ *if and only if* n *is a power of* 2.

Sketch of the proof.
$\boxed{\text{Modus Ponens}}$ For every $b \in \mathbb{Z}_2^n$,

$$\operatorname{Fin} n \ni i \mapsto f(b + \mathrm{e}_i) \in \operatorname{Fin} n \text{ is surjective,}$$

and thus also injective. Fix some $k \in \operatorname{Fin} n$ (e.g., $k = 0 < n$). Then

$$\begin{aligned} 2^n &= \sum_{b \in \mathbb{Z}_2^n} 1 = \sum_{b \in \mathbb{Z}_2^n} \sum_{i \in \operatorname{Fin} n} \delta(f(b + \mathrm{e}_i), k) = \sum_{i \in \operatorname{Fin} n} \sum_{b \in \mathbb{Z}_2^n} \delta(f(b + \mathrm{e}_i), k) \\ &= \sum_{i \in \operatorname{Fin} n} \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k) = n \cdot \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k). \end{aligned}$$

$\boxed{\text{Modus Ponens Reverse}}$ Let $n = 2^m$. Let $g : \operatorname{Fin}(2^m) \to \mathbb{Z}_2^m$ be a bijection. Define

$$f(b) := g^{-1}\left(\sum_{j \in \operatorname{Fin} 2^m} b_j \cdot g(j) \right), \quad b \in \mathbb{Z}_2^n.$$

Let $b \in \mathbb{Z}_2^n$ and $k \in \operatorname{Fin} n$.

$$f(b + \mathrm{e}_i) = k \Leftrightarrow \sum_{j \in \operatorname{Fin} 2^m} b_j \cdot g(j) + \sum_{j \in \operatorname{Fin} 2^m} \mathrm{e}_{ij} \cdot g(j) = g(k)$$

$$\Leftrightarrow \sum_{j \in \operatorname{Fin} 2^m} b_j \cdot g(j) + g(i) = g(k) \Leftrightarrow i = g^{-1}\left(g(k) - \sum_{j \in \operatorname{Fin} 2^m} b_j \cdot g(j) \right).$$

3. Lean tactics

- apply
- by_contra
- constructor
- dsimp
- exact
- have
- intro
- let
- obtain
- rw
- simp (only)
- unfold
- use

4. Function shift is injective

fun_shift_surj

Lemma 4.1. *If $f : \mathbb{Z}_2^n \to \mathrm{Fin}\, n$ is (SF) and $b \in \mathbb{Z}_2^n$, then the function $\mathrm{Fin}\, n \ni i \mapsto f(b + \mathrm{e}_i) \in \mathrm{Fin}\, n$ is surjective.*

Proof. $\mathrm{Fin}\, n \ni i \mapsto f(b + \mathrm{e}_i) \in \mathrm{Fin}\, n$ is surjective, by definition, if

$$\forall\, k \in \mathrm{Fin}\, n, \exists\, i \in \mathrm{Fin}\, n, f(b + \mathrm{e}_i) = k.$$

Taking our b in the definition of (SF), we get the desired conclusion. □

`Mathlib hint:` Function.Surjective.

fun_shift_inj

Lemma 4.2. *If $f : \mathbb{Z}_2^n \to \mathrm{Fin}\, n$ is (SF) and $b \in \mathbb{Z}_2^n$, then the function $\mathrm{Fin}\, n \ni i \mapsto f(b + \mathrm{e}_i) \in \mathrm{Fin}\, n$ is injective.*

Proof. Since $\mathrm{Fin}\, n \ni i \mapsto f(b+\mathrm{e}_i) \in \mathrm{Fin}\, n$ is a surjective function, by Lemma 4.1, on a finite set, it is also injective. □

`Mathlib hints:` Function.Injective, Finite.injective_iff_surjective.

5. Function shift sum

fun_shift_sum_one

Lemma 5.1. *If $f : \mathbb{Z}_2^n \to \mathrm{Fin}\, n$ is (SF), $b \in \mathbb{Z}_2^n$, and $k \in \mathrm{Fin}\, n$, then*

$$\sum_{i \in \mathrm{Fin}\, n} \delta(f(b + \mathrm{e}_i), k) = 1.$$

Proof. In view of the defintion of (SF) for f, we obtain $i_k \in \mathrm{Fin}\, n$ such that $f(b + \mathrm{e}_{i_k}) = k$.

We shall rewrite the sum by proving that only the i_k term in the sum is equal to 1, and the rest are 0.

Clearly, $\delta(f(b + \mathrm{e}_{i_k}), k) = 1$, by the definition of (d) and the logic of if-then-else.

Introduce $i \in \mathrm{Fin}\, n$ with $i \neq i_k$ and consider the definition of (d). Rewritting k as $f(b + \mathrm{e}_{i_k})$ and applying the logic of if-then-else in (d), we have to prove that $f(b + \mathrm{e}_i) \neq f(b + \mathrm{e}_{i_k})$. In order to get a contradiction, assume that $f(b + \mathrm{e}_i) = f(b + \mathrm{e}_{i_k})$. Then, by Lemma 4.2, we have $i = i_k$, which is false in view of the assumption $i \neq i_k$. □

`Mathlib hints:` rw [Fintype.sum_eq_single i_k], if_pos, if_neg.

fun_shift_sum_pow2

Lemma 5.2. *If $f : \mathbb{Z}_2^n \to \mathrm{Fin}\, n$ is (SF) and $k \in \mathrm{Fin}\, n$, then*

$$\sum_{b \in \mathbb{Z}_2^n} \sum_{i \in \mathrm{Fin}\, n} \delta(f(b + \mathrm{e}_i), k) = 2^n.$$

Proof. By Lemma 5.1, we have:

$$\forall\, b \in \mathbb{Z}_2^n, \sum_{i \in \mathrm{Fin}\, n} \delta(f(b + \mathrm{e}_i), k) = 1.$$

With only this relation, we can simplify $\sum_{b \in \mathbb{Z}_2^n} \sum_{i \in \mathrm{Fin}\, n} \delta(f(b + \mathrm{e}_i), k)$ to $\sum_{b \in \mathbb{Z}_2^n} 1$, which simplifies further to 2^n. □

fun_shift_sum_rev

Lemma 5.3. *If $f : \mathbb{Z}_2^n \to \operatorname{Fin} n$ and $k, i \in \operatorname{Fin} n$, then*

$$\sum_{b \in \mathbb{Z}_2^n} \delta(f(b + \mathrm{e}_i), k) = \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k).$$

Proof. Let $g : \mathbb{Z}_2^n \to \mathbb{Z}_2^n, g(b) = b + \mathrm{e}_i$. Then we have that g is bijective. Applying the fact that g is bijective to our finite sum, the equality holds, because $\sum_{b \in \mathbb{Z}_2^n} \delta(f(g(b)), k)$ is simply $\sum_{b \in \mathbb{Z}_2^n} \delta(f(b + \mathrm{e}_i), k)$. □

`Mathlib hints:` Function.Bijective, add_left_injective, add_right_surjective, Fintype.sum_bijective.

6. Modus Ponens

strategy_pow2

Lemma 6.1. *If $0 < n$ and $f : \mathbb{Z}_2^n \to \operatorname{Fin} n$ is (SF), then $\exists N \in \mathbb{N}, n \cdot N = 2^n$.*

Proof. Let $k := 0$ (which is in $\operatorname{Fin} n$ since $0 < n$). By Lemma 5.2, we have

$$\sum_{b \in \mathbb{Z}_2^n} \sum_{i \in \operatorname{Fin} n} \delta(f(b + \mathrm{e}_i), k) = 2^n.$$

Since the sums commute, we can rewrite the above expression as

$$\sum_{i \in \operatorname{Fin} n} \sum_{b \in \mathbb{Z}_2^n} \delta(f(b + \mathrm{e}_i), k) = 2^n.$$

Using Lemma 5.3, we simplify this to have

$$\sum_{i \in \operatorname{Fin} n} \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k) = n \cdot \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k) = 2^n.$$

We use $N := \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k)$ to finish the proof. □

`Mathlib hint:` Finset.sum_comm.

mul_pow2

Lemma 6.2. *If $\exists N \in \mathbb{N}, n \cdot N = 2^n$, then $\exists m \in \mathbb{N}, n = 2^m$.*

Proof. From the hypothesis, we obtain N such that $n \cdot N = 2^n$. So, we have n divides 2^n. Since 2 is prime, we obtain $m \in \mathbb{N}$ to be used in $n = 2^m$. □

`Mathlib hints:` Dvd.intro, Nat.prime_two, Nat.dvd_prime_pow.

puzzle_mp

Theorem 6.3. *If $0 < n$ and the puzzle has a strategy for n, then $\exists m \in \mathbb{N}, n = 2^m$.*

Proof. Since the puzzle has a strategy, we obtain f with (SF). By putting toghether Lemma 6.1 and Lemma 6.2, $\exists m \in \mathbb{N}, n = 2^m$. □

7. Modus Ponens Reverse

`bij_Fin`

Definition 7.1. For $m \in \mathbb{N}$, let $g : \mathrm{Fin}\,(2^m) \to \mathbb{Z}_2^m$ be a bijective function. g exists, applying the fact that $\mathrm{Fin}\,(2^m)$ and $\mathbb{Z}_2^m$ simply have equal cardinals.

`Mathlib hint:` Fintype.equivOfCardEq.

`sum_basis`

Lemma 7.2. *If $i \in \mathrm{Fin}\, n$ and $f : \mathrm{Fin}\, n \to \mathbb{Z}_2^m$, then* $\displaystyle\sum_{j \in \mathrm{Fin}\, n} e_{ij} \cdot f(j) = f(i)$.

Proof. By the definition of (e), $e_{ij} = 1$ if $i = j$ and $e_{ij} = 0$ otherwise. We rewrite $\sum_{j \in \mathrm{Fin}\, n} e_{ij} \cdot f(j)$ as $f(i)$, using the fact that the single term that contributes in the sum is the one with $j = i$. □

`Mathlib hint:` Fintype.sum_single_smul.

`puzzle_mpr`

Theorem 7.3. *If $n = 2^m$ for some m, then the puzzle has a strategy for n.*

Proof. First, we rewrite n as 2^m in (SF). Let g be given by Definition 7.1 and use f given by

$$f(b) := g^{-1}\left(\sum_{j \in \mathrm{Fin}\, 2^m} b_j \cdot g(j)\right).$$

Introduce $b \in \mathbb{Z}_2^n$ and $k \in \mathrm{Fin}\, n$ in order to verify (SF) for f. We want to show

$$\exists\, i \in \mathrm{Fin}\, n, f(b + \mathrm{e}_i) = g^{-1}\left(\sum_{j \in \mathrm{Fin}\, 2^m} (b_j + e_{ij}) \cdot g(j)\right) = k.$$

We simplify the above relation using the distributivity of the scalar multiplication over addition and over sum to get

$$\sum_{j \in \mathrm{Fin}\, 2^m} (b_j + e_{ij}) \cdot g(j) = \sum_{j \in \mathrm{Fin}\, 2^m} b_j \cdot g(j) + \sum_{j \in \mathrm{Fin}\, 2^m} e_{ij} \cdot g(j).$$

Let's use $i := g^{-1}\left(g(k) - \sum_{j \in \mathrm{Fin}\, 2^m} b_j \cdot g(j)\right)$. By Lemma 7.2, we can rewrite the desired relation with $\displaystyle\sum_{j \in \mathrm{Fin}\, 2^m} e_{ij} \cdot g(j) = g(i)$. By the expression of i, we can simplify to get

$$\begin{aligned} & g^{-1}\left(\sum_{j \in \mathrm{Fin}\, 2^m} b_j \cdot g(j) + g(i)\right) \\ & = g^{-1}\left(\sum_{j \in \mathrm{Fin}\, 2^m} b_j \cdot g(j) + g(k) - \sum_{j \in \mathrm{Fin}\, 2^m} b_j \cdot g(j)\right) = g^{-1}(g(k)) = k. \end{aligned}$$

□

`Mathlib hints:` g.symm, add_smul, Finset.sum_add_distrib.

8. Finish

puzzle

Theorem 8.1. *If* $0 < n$*, then the puzzle has a strategy for* n *if and only if* $\exists\, m \in \mathbb{N}, n = 2^m$.

Proof. The forward direction is Theorem 6.3, and the reverse direction is Theorem 7.3. $\square$